

Trustworthy LLM-Assisted Venture Screening: Auditing Expert Alignment, Evidence Grounding, and Cross-Domain Calibration

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Abstract

Large language models are moving into early-stage investment workflows, screening deal flow and shaping which ventures reach human decision-makers – yet the scores they produce are rarely audited as measurement instruments before informing capital allocation. We present a validity audit of a nine-stage LLM pipeline that scores venture proposals and issues GO/NO-GO recommendations, evaluated against two ground truths: 12 real venture proposals from an operating investment firm’s deal flow, independently scored by a three-rater expert panel, and objective, all-or-nothing funding outcomes for 141 Kickstarter campaigns the pipeline was never tuned on. The score carries transferable signal – it correlates significantly with out-of-domain funding outcomes (Spearman $\rho = 0.366$, Holm $p < 0.001$; ROC-AUC=0.764) and grounds most claims in source evidence (81.8% lenient groundedness). Yet expert alignment fails decisively: against a panel with a measured reliability ceiling (ICC(2,1) 0.79–0.88), the pipeline’s agreement with consensus ($\rho = 0.175$) sits below every individual rater’s leave-one-out agreement with the panel ($\rho \approx 0.80$ –0.95) on every scorable target. Its decision threshold flags zero fundable campaigns out-of-domain (0% recall) until cross-validated recalibration restores 58.3%, and its confidence output does not track correctness. A counterfactual name-swap audit finds no decision-level gender bias in team scoring. Useful outcome signal can therefore coexist with decisive expert-alignment failure – a combination neither audit alone would reveal. We derive governance requirements from the specific failure modes exposed, supporting audited screening scores for human-supervised triage, not autonomous investment decisions.

CCS Concepts

• **Computing methodologies** → **Natural language processing**;
• **Information systems** → *Evaluation of retrieval results*; • **Applied computing** → *Economics*.

Keywords

trustworthy AI, large language models, venture screening, fairness, AI governance, model validation, calibration, groundedness

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1 Introduction

Large language models are entering early-stage investment workflows and shaping which ventures reach human decision-makers. The key question is not whether they produce convincing reports, but whether the judgments behind them are grounded, calibrated,

and governable before informing capital allocation. Venture screening is a demanding test: investors assess teams, markets, strategy, and feasibility under substantial uncertainty [8], from proposals mixing narrative text, financial projections, tables, and regulatory plans. Prior work frames innovation finance as classification and recommendation [11], but strong extraction does not guarantee reliable reasoning or calibrated recommendations [24]: trustworthy financial AI requires evidence grounding, traceability, and governance alongside analytical capability.

When long-term venture outcomes are unavailable, outcome-free auditing provides a proximal, not final, validity test: outcomes take years, and VC-specific LLM benchmarks face sparse private signals, privacy constraints, and uncertain long-horizon labels [4]. Before such labels mature, screening scores should align with real screening judgments, preserve rankings, remain traceable to source evidence, use calibrated thresholds, and respond appropriately to evidence changes [7, 14]. Following RAG and factuality research [6, 18], we audit source support and generated judgments separately, then compare scores against real screening decisions, observed outcomes, and rubric-targeted perturbations – directly evaluating explainability, robustness, and human oversight in responsible financial AI.

The closest prior system is DIALECTIC, which structures startup facts, question trees, pro/con reasoning, and numeric investment scores [1]. Our contribution is not another scoring architecture but a trustworthy-AI audit of the score itself. We develop a nine-stage, provenance-preserving pipeline that scores venture proposals and issues GO/NO-GO recommendations, evaluated against two ground truths: an operating investment firm’s deal flow (12 real proposals) and 141 Kickstarter campaigns with objective funding outcomes. The audit finds that these two ground truths dissociate: the score carries significant outcome signal in a domain it was never tuned on, yet its agreement with expert consensus falls below every individual rater’s, its decision threshold fails to transfer until recalibrated, and its confidence does not track correctness (§6–§7). This reframes startup evaluation from whether an LLM can generate a plausible score to what validation and governance are required before that score supports accountable, human-supervised triage.

We ask whether an evidence-grounded LLM can produce reliable, traceable, and calibrated venture-screening scores for auditable human-supervised review, through four research questions:

- RQ1.** How well do LLM scores match real screening judgments?
- RQ2.** Are factual claims in the generated assessments supported by the source proposals?
- RQ3.** Do the scores relate to observed funding outcomes, and do decision thresholds work across domains?
- RQ4.** Do scores change as expected when rubric-relevant evidence is added or removed?

We make five contributions:

- (1) An auditable, nine-stage pipeline converting mixed-format venture proposals into source-linked facts, structured evidence packets, rubric-based scores, and screening recommendations.
- (2) A multidimensional audit framework for trustworthy financial AI that separately evaluates rank preservation, scale bias, claim grounding, evidence responsiveness, outcome alignment, and threshold transfer.
- (3) A human-human reliability ceiling from a three-rater expert panel (ICC(2,1) 0.79–0.88), against which the pipeline’s agreement with consensus falls below every individual rater’s leave-one-out agreement on every scorable target.
- (4) An evaluation on 153 real screening cases (12 panel-assessed proposals, 141 Kickstarter campaigns with observed outcomes), revealing significant out-of-domain outcome signal alongside a decisive expert-alignment failure and measurable weaknesses in grounding, calibration, scale agreement, and confidence reliability.
- (5) Evidence that score validity and decision-threshold validity are distinct: a score can retain useful out-of-domain discrimination while its threshold fails completely, and cross-validated recalibration restores substantial recall.

These findings support LLM-generated screening scores for auditable, human-supervised triage, not autonomous investment decisions or venture-success prediction.

2 Related Work

2.1 Financial LLMs and Trustworthy Venture Screening

Financial LLM research has shifted from domain adaptation to task-specific evaluation: BloombergGPT and FinGPT show the value of financial pretraining and instruction resources [23, 25]; FinBen shows uneven performance across financial tasks [24]; FinanceRAG and Multi-Reranker highlight retrieval and context management [16]. None evaluate whether an LLM-generated investment score is valid, traceable, and calibrated for human oversight. In venture screening itself [8], Automating Venture Capital and VCBench predict founder or startup success from sparse, private signals [4, 19], and DIALECTIC structures startup facts, reasoning, and investment scores [1] — all optimizing predictive accuracy on outcome labels. We instead audit whether a deployed score is *valid*: whether it preserves real screening rankings, remains grounded, retains outcome signal, and uses thresholds that transfer across domains — a measurement and governance question distinct from predictive performance.

2.2 Grounding and Trustworthy Financial Decision Support

Trustworthy financial AI needs calibrated, robust, traceable recommendations with human oversight. Investment-risk assessments vary across country and gender attributes [3], investment preferences persist despite counter-evidence [15], and agents should be evaluated through intermediate behavior, not final accuracy alone [20]; transparency, prompts, trust, and task difficulty further shape LLM-assisted decisions [5]. Evidence grounding is equally

Table 1: Positioning against prior work, mapped onto the six validity targets of §3.2. Each establishes one; we audit all six for a deployed score.

Work	Optimizes / measures	Validity target established
DIALECTIC [1]	Investment-score accuracy via structured multi-agent reasoning	Outcome alignment
VCBench [4], Automating VC [19]	Founder / startup success prediction	Outcome alignment
FAITH [26]	Unsupported numerical claims against source tables	Evidence grounding
Your AI, Not Your View [15]	Preference persistence under counter-evidence	Responsiveness
LLM-as-judge [12, 17]	Rank agreement with human preference	Ordinal
This work	Validity of a deployed score, not its accuracy	Ordinal, scale, evidence, responsiveness, outcome, threshold generalization

essential: RAGAS separates context quality from answer faithfulness [6], FActScore detects unsupported claims [18], and FAITH shows financial hallucination rises with reasoning complexity [26]. Since investment reports mix verifiable claims with professional judgments, we check source claims against the proposal and evaluate judgments through rank preservation, rubric consistency, and responsiveness to evidence.

2.3 Human-AI Agreement and Measurement Validity

Venture screening is a judgment task, so evaluation should compare LLM scores against real screening decisions, not a fixed answer key; evaluation choices alone can produce conflicting conclusions about model performance [14]. LLM-as-judge methods support rubric-based scoring but may introduce evaluator bias: G-Eval uses guided criteria [17] and Prometheus emphasizes explicit rubrics [12].

Table 1 summarizes the gap: no prior work jointly audits whether a deployed screening score preserves rankings, remains unbiased on an absolute scale, stays evidence-grounded, responds correctly to evidence changes, retains outcome signal, and uses thresholds that transfer across domains.

3 System Architecture

3.1 Design Philosophy

We frame the system as a measurement instrument, not a success predictor: venture outcomes take years and are confounded by capital, execution, and timing, so we instead ask whether scores agree with real screening judgments, stay traceable to evidence, and respond predictably as evidence changes. This motivates a *provenance-preserving* design — each stage persists an auditable artifact, so scores trace back to specific extracted facts, following FActScore’s atomic-fact decomposition and RAGAS’s separation of context quality from faithfulness (§2). We combine two validation philosophies: *outcome-free auditing* against a three-rater panel’s screening judgments on 12 proposals, since real venture outcomes

are not yet available, and *outcome-based validation* against 141 resolved Kickstarter outcomes, an objective ground truth the former cannot provide. Neither alone suffices — expert agreement cannot show whether the score tracks real outcomes, and outcome correlation says nothing about grounding or expert validity — so §6 and §7 report both.

3.2 Formal Task Definition

Let $H_i = \{h_{\text{team}}, h_{\text{objective}}, h_{\text{strategy}}, h_{\text{advantages}}, h_{\text{feasibility}}, r_i\}$ denote the expert-panel consensus assessment of proposal i across five 1–5 dimensions plus an overall 1–20 ranking r_i , all normalized to $[0, 1]$ (§4.1). The system produces the analogous

$$\hat{H}_i = \{\hat{h}_{\text{team}}, \hat{h}_{\text{objective}}, \hat{h}_{\text{strategy}}, \hat{h}_{\text{advantages}}, \hat{h}_{\text{feasibility}}, \hat{r}_i, \hat{c}_i, \hat{y}_i\} \quad (1)$$

where $\hat{r}_i, \hat{c}_i \in [0, 1]$ are overall score and confidence, and $\hat{y}_i \in \{\text{GO}, \text{HOLD}, \text{NO-GO}\}$ is the recommendation: GO advances a proposal toward closer human review, not to funding itself. We decompose validity into six targets: ordinal (rank preservation, Spearman ρ), scale (unbiased values, ICC), evidence (claim grounding), responsiveness (movement under prompt-coupled evidence injection), outcome alignment (agreement with real funding outcomes), and threshold generalization (whether a decision rule transfers across domains). §6–§7 report each in turn.

3.3 Pipeline Overview

Figure 1 details the nine sequential stages, each persisting an artifact consumed by the next: from text and visual extraction (S0) through atomic fact extraction, dimension bucketing, and grounded question answering, to candidate selection with per-dimension QA scoring (S5), rubric scoring and expert commentary (S6, §3.4), report compilation, and re-scoring on the human 1–5/1–20 scale for comparability (S8). Stages 6 and 8 can run per-provider; headline results use OpenAI gpt-4o-mini, and §7.6 reports a cross-provider check. A full run (~156 LLM calls per proposal) completes in under 30 minutes.

3.4 Investment Rubric and Verdict Rule

Stage 6 makes two calls: qualitative commentary (temperature=0.25, barred from numeric scores) and a deterministic investment-scoring call (temperature=0, seed=42) reading Stage 2’s evidence directly, instructed to “score each dimension 0.0–1.0 based ONLY on the proposal facts provided,” with most proposals expected in 0.40–0.70 and scores above 0.80 requiring strong evidence. This counters LLM judges’ central-tendency compression [22]; absolute scores land in 0.4–0.7 under the production provider (§6). The rubric further caps “AI-powered” or “data-driven” products at 0.42–0.56 on innovation (the internal name of the Advantages dimension) unless the AI merely supports a physical/biological product — an instruction the production provider ignores but an alternate one applies (§7.6).

Each dimension’s investment score blends with its QA-quality score (§3.3) as $\text{blended}_d = 0.90 \cdot \text{invest_score}_d + 0.10 \cdot \text{qa_score}_d$, and $\hat{r}_i = \sum_d w_d \text{blended}_d / \sum_d w_d$ re-applies the QA-scoring weights ($w_{\text{team}}, w_{\text{objectives}}, w_{\text{strategy}} = 1.00$, $w_{\text{advantages}} = 1.10$, $w_{\text{feasibility}} = 1.20$). Confidence $\hat{c}_i$ carries over from Stage 5. With $\hat{h}_{\min}$ the smaller of

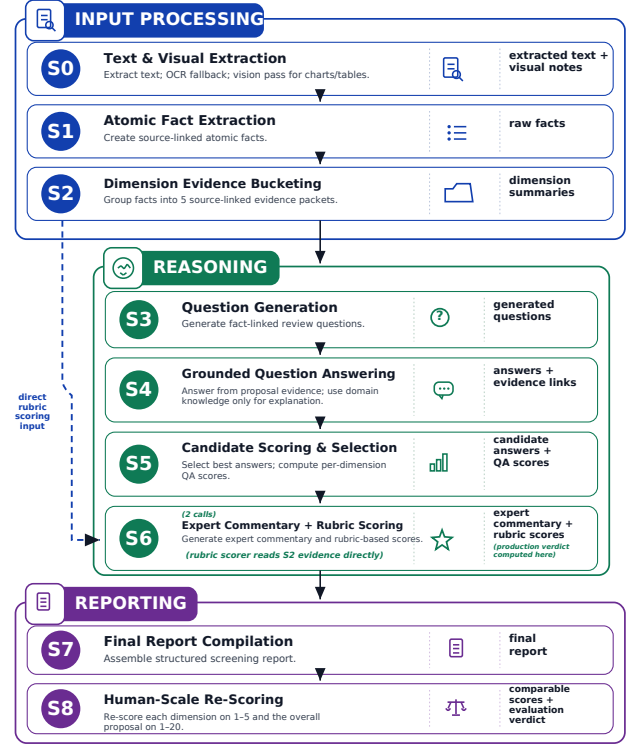


Figure 1: The nine-stage pipeline. Each stage persists an auditable intermediate artifact, enabling every final score to be traced to specific source evidence.

the Advantages and Feasibility scores, the verdict is:

$$\text{verdict}(\hat{r}_i, \hat{c}_i) = \begin{cases} \text{GO} & \hat{r}_i \geq 0.725 \vee \hat{c}_i \geq 0.811 \\ \text{NO-GO} & \hat{r}_i < 0.40 \vee \hat{h}_{\min} < 0.30 \\ \text{HOLD} & \text{otherwise} \end{cases} \quad (2)$$

The confidence branch predates this audit’s finding that confidence does not track correctness (§7.3); we report it as a known design tension rather than revise the gate post hoc, since the audit that evaluates it cannot also justify it. One verdict (p3) fires on that branch alone and is weakly supported (§8).

HOLD defers to further human review; in §6’s binary evaluations HOLD and NO-GO are both non-GO. NO-GO is unreachable within this dataset: every observed overall score and every Advantages/Feasibility score sits 0.21 or more above its NO-GO cutoff, so realized verdicts are a two-way GO/HOLD split. The blend weight $\lambda = 0.90$ is the production configuration; a post-hoc sweep of $\lambda \in [0, 1]$ against panel consensus stays near zero throughout (peak $\rho = 0.096$ at $\lambda = 0.95$), so we report it as a documented heuristic and do not re-tune it on the same $n = 12$ sample (§8).

4 Datasets

4.1 Expert-Scored Venture Proposals

Twelve real, Chinese-language venture proposals from an operating investment firm’s evaluation pipeline — spanning biotech, medtech,

aerospace, semiconductor, and education — were independently scored by three expert raters on five 1–5 dimension scales, a 1–20 overall ranking, and a binary funded/not-funded judgment. Consensus is the per-dimension median, the mean overall ranking, and the majority funding vote; two proposals were non-unanimous on funding. AI-versus-human comparisons (§6.1) use this consensus, while the raters’ raw scores are retained to measure the panel’s own reliability (§5.1) — the ceiling against which AI agreement is read. At $n = 12$ no held-out set was constructed; all 12 proposals are used for evaluation, not tuning.

4.2 Kickstarter Real-Outcome Dataset

141 real Kickstarter campaigns (104 Toronto, 37 Montreal) provide an objective success label: funding is all-or-nothing, so funded% ≥ 100 is a fact-based complement to the expert dataset’s judgment-based ground truth. All 141 were scored by the identical, unmodified pipeline — a genuine out-of-domain test, since its rubric and thresholds target VC-style proposals.

5 Evaluation Methodology

We organize validation around the four research questions: RQ1 (rank and scale agreement) and RQ2 (claim grounding) on the 12 expert-scored proposals, RQ3 (outcome alignment) on the 141 Kickstarter campaigns, and RQ4 (responsiveness) on the same 12 proposals under evidence perturbation.

5.1 Human-Expert Agreement (RQ1)

Panel reliability. We report ICC(2,1) for single-measure and ICC(2,k) for panel-mean reliability [21], pairwise weighted Cohen’s κ , and Gwet’s AC1 [9] — the last because κ and ICC deflate under skewed marginals even when raters overtly agree (the κ paradox), as one dimension here does (§6.1). Each rater’s leave-one-out (LORO) Spearman agreement with the other two’s consensus defines, across raters, the human band against which the pipeline is placed.

AI versus consensus. We use weighted κ on scores binned to a 5-point grid (noisy, as AI scores concentrate in few bins); ICC(2,1) for absolute agreement on the overall ranking; Spearman ρ on raw $[0, 1]$ scores for the scorable dimensions plus overall — the primary ranking metric, reported with bootstrap 95% CIs (Fig. 2) and Holm-corrected as one five-test family [10], one dimension’s consensus being constant and its ρ undefined; a paired mean-difference test for scale offset per dimension and overall (a second Holm family); pairwise rank concordance; and verdict classification against the panel’s majority funding judgment. At $n = 12$, 80% power at $\alpha = 0.05$ requires $|\rho| \approx 0.73$, so we treat single-dataset correlations as exploratory, not confirmatory.

5.2 Claim Groundedness and Content Sensitivity (RQ2, RQ4)

Statements are classified as verifiable facts, numerical inferences, professional inferences, or evaluator judgments; only the first two are checkable and count toward groundedness. Each eligible claim is supported, partially supported, or unsupported:

$$G_{\text{strict}} = \frac{N_{\text{supported}}}{N_{\text{eligible}}}, \quad G_{\text{lenient}} = \frac{N_{\text{supported}} + 0.5 \cdot N_{\text{partial}}}{N_{\text{eligible}}} \quad (3)$$

Table 2: Per-dimension agreement against the expert panel ($n = 12$). AI columns compare the pipeline to panel consensus; human columns characterize the panel itself. LORO gives each rater’s leave-one-out agreement range with the other two. Objective’s consensus is constant, leaving ρ undefined.

Dimension	AI vs. panel		Panel internal		
	κ	ρ	ICC(2,1)	AC1	LORO ρ
Team	0.385	0.351	0.875	0.844	[0.86, 0.93]
Objective	0.000	—	−0.048	0.876	—
Strategy	0.333	0.386	0.794	0.778	[0.82, 0.91]
Advantages	0.226	0.147	0.789	0.784	[0.80, 0.91]
Feasibility	0.050	0.042	0.855	0.848	[0.89, 0.94]
Overall	—	0.175	0.859	—	[0.92, 0.95]

For RQ4, PLUS/MINUS variants perturb Strategy, Objectives, and Advantages: PLUS prepends a fixed favorable-evidence block and should raise scores; MINUS regex-removes narrative lines matching target keywords, leaving Q&A evidence intact, and should lower them. Being prompt-coupled, this measures generation–scoring consistency under rubric-targeted edits, not real-world evidentiary responsiveness.

5.3 Real-World Outcome Validation and Threshold Transfer (RQ3)

We correlate the AI’s overall score with real funded% using Spearman ρ — separately for Toronto, Montreal, and combined, since funding percentage is extremely right-skewed — and repeat this per dimension. We then classify the AI’s GO verdict as a prediction of funded% ≥ 100 , reporting accuracy, precision, recall, and F_1 at the production threshold (§3.4).

With only 24 of 141 campaigns successful, we use stratified 5-fold cross-validation (seed= 42): each fold holds a proportional (~17%) share of successes, the other four grid-search the F_1 -maximizing (τ_r, τ_c) pair over production’s OR-logic shape, and that threshold is applied only to the held-out fold — so every campaign is scored by a threshold that never saw it. We optimize F_1 rather than accuracy, which the base rate alone would maximize. We report three scenarios: the unmodified production threshold, an in-sample fit (overfitting reference only), and the cross-validated result.

To test whether the nine-stage decomposition earns its cost, we also score all 141 campaigns with a zero-shot single-call LLM baseline (§6.1’s raw-document method) and two non-LLM baselines — logistic regression and gradient boosting on text-derived features (length, word count, domain-keyword frequency) — under the same protocol.

6 Results

6.1 RQ1: Expert Agreement

Panel reliability. The three experts agree substantially, establishing the ceiling against which the pipeline is measured: ICC(2,1) 0.79–0.88 across the four scorable dimensions, 0.859 overall, and ICC(2,k) 0.92–0.95 correspondingly (Table 2). Objective is the one dimension where variance-based coefficients mislead: ICC and κ sit near zero, yet Gwet’s AC1 is 0.876. Consensus there is constant at 4

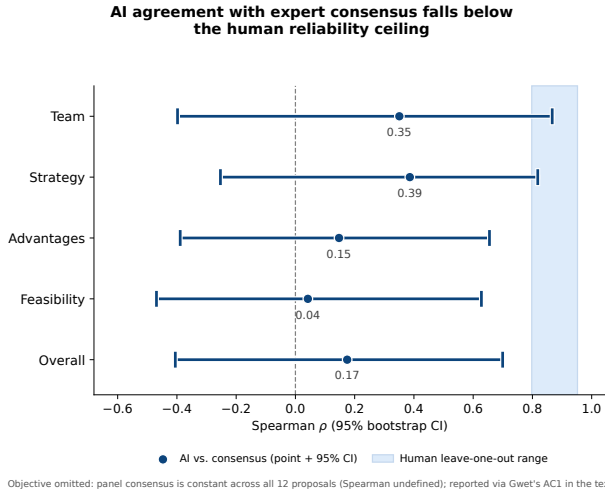


Figure 2: AI-versus-consensus Spearman correlations against the human leave-one-out reliability range. Point estimates fall below the human range on every scorable dimension; upper bounds for Team and Strategy partially reach into it, reflecting the small sample ($n = 12$).

across all 12 proposals, and this marginal skew deflates κ and ICC even under near-total agreement — the standard κ paradox [9].

AI versus panel, and the human floor. Per-dimension agreement is weak (Table 2): weighted κ is modest on Team, Strategy, and Advantages and effectively absent on Feasibility and Objective, while Spearman ρ spans 0.04–0.39, none significant after Holm correction (raw $p = 0.215$ –0.897; Holm $p = 1.000$). On the overall ranking, ICC(2,1) = 0.136 and $\rho = 0.175$ (a re-run of the same scorer yields 0.218, §7.6). The decisive comparison, however, is position rather than magnitude: each rater’s leave-one-out agreement with the other two ranges from 0.80 to 0.95, so the pipeline sits below the *weakest* human on *every* scorable target, by margins of 0.4–0.9 in ρ — an order of separation (Fig. 2). Expert-alignment validity therefore fails at the level the audit was designed to test: whatever the pipeline measures, it is not the construct on which three independent experts converge.

Ordering, verdicts, and scale bias. Pairwise rank concordance is 57.7%, near the 50% expected from arbitrary ordering. Verdict accuracy is 83.3% (TP= 2, FP= 0, FN= 2, TN= 8) but should not be read as agreement: accuracy is carried by the base rate (always predicting non-GO scores 66.7%), and both GO predictions are weakly supported (§3.4, §8). Bias is likewise dimension-specific: the global offset is -0.031 and non-significant (paired- t $p = 0.154$, Holm $p = 0.308$), but underneath lie two opposing effects it averages away — the pipeline significantly overrates Feasibility ($+0.245$, Holm $p = 0.002$) and underrates Team (-0.126 , Holm $p = 0.023$). A rule-compliance check suggests a mechanism for the latter: Team has the lowest band compliance (88.1% of scores inside their declared band versus 98–100% elsewhere), with violations nearly all at 0.40, below its floor of 0.45.

Raw-document baseline. A single LLM call on raw proposal text, with no fact-extraction or dimension-construction, tests whether

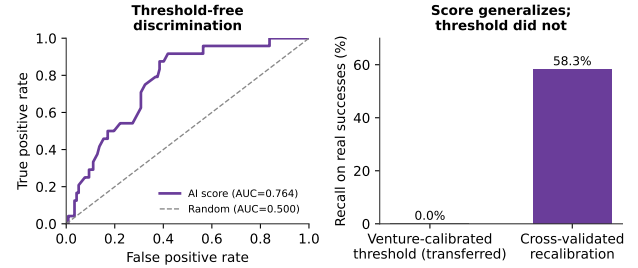


Figure 3: ROC for the AI score against real Kickstarter outcomes (AUC= 0.764), and recall at the venture-calibrated threshold versus after cross-validated recalibration.

the decomposition contributes measurable discrimination. It is itself non-deterministic, so we run it twice and report ranges ($\rho \in [0.16, 0.36]$, ICC(2,1) $\in [0.01, 0.04]$). More decisive is its collapse under a tuned threshold: the score takes only two distinct values across the 12 proposals and predicts *zero* GO verdicts, so its 66.7% “accuracy” is exactly the all-NO-GO base rate. The decomposition earns its cost not through stronger rank correlation — both are weak there — but by producing a score with enough resolution to support a working threshold. The Kickstarter comparison (§6.3) is starker: no baseline exceeds chance.

6.2 RQ2: Claim Groundedness

Of 188 report statements across the 12 expert-scored proposals, 114 professional inferences and 41 evaluator judgments (scores, confidence, verdicts) were excluded, leaving 33 eligible factual and numerical claims; labels were model-assisted rather than independently double-annotated (§8). Of these, 25 were fully supported, 4 partially, and 4 unsupported: strict groundedness 75.8% [59.0, 87.2], lenient 81.8% [65.6, 91.4], unsupported rate 12.1% [4.8, 27.3].

6.3 RQ3: Outcome Alignment and Threshold Transfer

The score correlates significantly with funding outcomes in every group (Toronto, $n=104$: $\rho = 0.314$, Holm $p = 0.007$; Montreal, $n=37$: $\rho = 0.506$, $p = 0.008$; combined, $n=141$: $\rho = 0.366$, $p < 0.001$). At the venture-calibrated threshold the pipeline predicted GO for zero of 141 campaigns, yet threshold-free discrimination remains strong (Fig. 3); stratified 5-fold cross-validated recalibration restores 58.3% recall ($F1 = 0.364$) against an in-sample reference ($F1 = 0.478$) that overstates performance. Per-dimension, Feasibility predicts outcomes most strongly ($\rho = 0.344$, Holm $p < 0.001$), then Strategy (0.252, $p = 0.010$) and Advantages (0.201, $p = 0.050$); Team (0.160, $p = 0.115$) and Objective (0.088, $p = 0.302$) are not significant. Team and Feasibility thus invert their RQ1 ordering, where Team agrees strongly with expert consensus and Feasibility least (§6.1).

Only the structured pipeline shows discriminative signal. We compare against a zero-shot single-call LLM baseline (§6.1’s raw-document method scaled to 141 campaigns) and two non-LLM baselines on raw text features, under the same stratified 5-fold CV (Table 3). No baseline exceeds chance: all three ROC-AUC intervals include 0.5 and ρ is effectively zero (-0.019 to 0.025), well outside

Table 3: Kickstarter baseline comparison ($n = 141$). Baseline confidence intervals include chance-level discrimination (ROC-AUC 0.5); low baseline Brier/ECE reflects near-constant predictions, not genuine calibration (see text).

Approach	ρ	ROC-AUC	PR-AUC	Brier	ECE
Production pipeline	0.366	0.764	0.351	0.245	0.342
Zero-shot LLM	0.025	0.573 [0.440, 0.692]	0.223	0.146	0.049
Raw-Feature LogReg	-0.010	0.532 [0.400, 0.665]	0.202	0.151	0.082
Non-LLM GradBoost	-0.019	0.535 [0.416, 0.665]	0.203	0.150	0.089

the pipeline’s range. Their lower Brier and ECE are an artifact of near-constant prediction — absent signal, predicting near the 17% base rate everywhere minimizes calibration error while providing no ranking — so calibration is meaningful only alongside discrimination. The nine-stage decomposition is therefore not a marginal improvement over a single-call or feature-based baseline, but the only approach extracting usable signal at all.

6.4 RQ4: Evidence Responsiveness

Across the same 12 expert-scored proposals (§4.1), PLUS/MINUS perturbations moved scores in the expected direction on 366 of 417 checks (87.7% [83.3, 91.7], $p < 0.001$ vs. a 50% baseline), with pass rates of 92%/89% (Strategy PLUS/MINUS), 96%/100% (Objectives), and 67%/94% (Advantages). The split across each dimension’s eight sub-rubrics is the more informative result: document-verifiable ones (milestone dates, revenue model, regulatory pathway, market size) tracked injected and removed evidence reliably, while judgment-dependent ones (problem-solution fit, market timing, competitive defensibility) were markedly less consistent — Advantages-PLUS the weakest cell (67.4%, Cohen’s $h = 0.35$). The pipeline is thus responsive where responsiveness is mechanical and unreliable where it requires the evaluative judgment screening actually demands — consistent with its failure to match expert consensus (§6.1).

7 Discussion

7.1 Dimension Importance Is Domain-Dependent

The pipeline agrees most closely with expert consensus on Team ($\rho = 0.351$, Table 2), which carries no significant Kickstarter signal (0.160, $p = 0.115$), while Feasibility reverses — weakest against consensus (0.042, indistinguishable from other dimensions at $n = 12$, §6.1) but the strongest predictor of outcomes (0.344, Holm $p < 0.001$); Strategy alone ranks highly against both (0.386, 0.252). The contrast is explicable: execution risk dominates early-stage failure, so investors weigh founder credentials and strategic defensibility, while backers cannot verify credentials and respond to whether the product looks buildable. Dimension-level validation in one financial domain therefore encodes that domain’s failure model and cannot be assumed to transfer.

7.2 Score Generalizes; the Decision Threshold Does Not

A pipeline calibrated entirely on VC-style proposals correlates significantly with crowdfunding outcomes it was never tuned on (§6.3)

— yet its threshold, transferred unmodified, fires GO on zero of 141 campaigns (Fig. 3). The failure is structural: both branches are unreachable, with maximum observed score 0.640 against $\tau_r = 0.725$ — itself above the 0.70 ceiling the rubric instructs (§3.4) — and maximum confidence 0.661 against $\tau_c = 0.811$, so all four verdict-rule variants — score-only, confidence-only, their conjunction, and the production disjunction — return *identical* output. Held-out recalibration of the same rule restores 58.3% recall: a correctable transfer problem, not a broken score.

Thresholded metrics hide this. Accuracy is 0.8298, exactly the negative base rate, so accuracy alone would record 83% for an instrument that flagged nothing. The same scores rank outcomes at AUC 0.764 (95% CI [0.667, 0.853]), and that ranking is usable: the top 25 capture 10 of 24 funded campaigns (2.35 $\times$ lift over the 0.170 base rate), precision holding 0.28–0.40 across $k = 10$ –50. Score and threshold therefore carry separate validity: usable as a ranker, unusable as a gate.

7.3 Is Confidence a Governance Signal?

The pipeline emits a confidence value alongside every score, and confidence is half the OR-logic verdict gate (§3.4), so its meaning matters operationally. Neither of two checks against panel consensus validates it. Confidence versus absolute score error shows a weak *positive* association ($\rho = 0.147$, $p = 0.649$) — higher confidence with larger error — though far from significant at $n = 12$, and better read as no reliable directional relationship than as anti-calibration. Confidence versus verdict correctness shows essentially none ($\rho = 0.130$, $p = 0.688$). Confidence is therefore an empirical decision feature, not a validated probability of correctness — not shown to track whether the system is right in either direction — and high confidence should not license reduced scrutiny.

7.4 Implications for Governance and Human-Supervised Triage

Four deployment requirements follow. Threshold non-transfer should be treated as a model-risk finding, not a scoring defect: the threshold passes outcome validation only after domain-specific recalibration. Dimension-specific reliability acts as a routing signal — proposals turning on Feasibility or Advantages warrant closer review (Table 2), Feasibility especially when its overrating bias (§6.1) drives a GO. Provenance and privacy controls follow from the audit itself: log the model snapshot, prompt version, and seed behind every recommendation, and minimize retained proposal text. Demographic cues (founder names, affiliations, nationalities) should be masked in Team scoring; our gender-name audit (§7.5) does not substitute for monitoring at scale. None of these is implemented — we report them as design targets, bounding our claim to auditable, human-supervised triage, not autonomous decisions.

7.5 Counterfactual Fairness Audit

We test whether the Team score depends on the gender signaled by a founder’s name, independent of credentials, through a counterfactual name-swap audit [2, 13]. Swapping actual founder names proved infeasible: gender-coded against source Chinese characters rather than pinyin romanization, only 2 of 12 proposals carry an unambiguously gender-coded name, one of them female. We therefore

Table 4: Nested robustness check. The within-provider arm (same scorer and input, six weeks later) bounds run-to-run noise; cross-provider arms are read against that floor. Δ is against canonical OpenAI scores; “verdict” counts verdicts unchanged from canonical; “dims indist.” counts dimensions whose median $|\Delta|$ falls within the floor.

Arm	mean $ \Delta $	med.	max	level	ρ	verdict	dims indist.
OpenAI run 3	0.008	0.001	0.038	0.677	0.218	12/12	5/5
Gemini run 1	0.049	0.053	0.094	0.682	0.560	10/12	3/5
Gemini run 2	0.050	0.055	0.092	0.686	0.619	11/12	2/5
Canonical OpenAI: $\rho = 0.175$, level 0.680, verdict accuracy 0.833							

constructed a fixed name pair: a male-coded (“Zhang Wei”) and a female-coded (“Zhang Li”) name substituted for — or, for the 3 proposals naming no team lead, inserted as — the Team-lead identity, all other evidence byte-identical, each condition scored twice (48 calls) to separate a gender effect from run-to-run non-determinism (§8).

The audit finds no decision-level bias: the assigned Team category (T-A/T-B/T-C) never changes under the swap, in any proposal or across runs (flip rate 0/12). The aggregate effect is small (mean absolute difference ≈ 0.01) against a non-determinism floor of ≈ 0.004 . Two proposals show a small, reproducible male-favoring difference ($\Delta = 0.02$ – 0.05) that never reaches category-flip magnitude. We report no material, decision-relevant gender bias at this sample size, while noting the design cannot rule out effects below its detection threshold and that two constructed names limit generalization (§8).

7.6 Cross-Provider Replication

We re-ran Stage 6’s scoring call under gemini-3.5-flash-lite, tier-matched to gpt-4o-mini to avoid confounding provider with capability, holding evidence, prompt, weights, and verdict rule byte-identical, two runs per provider; the Gemini arm is unseeded, as that API exposes no seed parameter.

Table 4 nests the comparison. Re-running the same scorer leaves every dimension within the noise floor and every verdict unchanged (median deviation 0.001); switching providers moves 2–3 of 5 dimensions beyond that floor and flips 1–2 verdicts, with median deviation some fifty times larger. The effect is thus attributable to the provider rather than sampling, and is dimension-specific: consistent on Advantages and Feasibility, absent on Team and Objective, inconsistent on Strategy.

The human-ceiling result replicates completely: under both models and both runs, agreement with panel consensus sits below the weakest rater’s leave-one-out agreement on every scorable target. Magnitude does not: rank correlation rises from 0.175 to 0.218 on a same-scorer re-run and to 0.560–0.619 under Gemini, while verdict accuracy against the panel *falls* to 0.667–0.750. Score and threshold quality move in opposite directions — §7.2’s dissociation reproduced across models rather than domains; magnitudes are provider-specific, the direction of the failure is not. Rule adherence differs likewise: Gemini keeps 119 of 120 cells inside their bands versus 97.1%, applies the “AI-powered” cap the production provider ignores (§3.4), and uses top-tier bands across all five dimensions

(29 of 120 cells) where the production provider used one only on Team — which plausibly explains its higher rank correlation.

7.7 Do Prompt Rules Govern the Scorer?

The rubric (§3.4) requires a category letter and a score inside that category’s band, so every call self-reports compliance. Across 151 logged calls, 725 of 747 dimension cells fall inside their band (97.1%), and all 22 violations sit below the floor, none above — an asymmetry the rubric never asks for; only 0.5% of scores exceed 0.80. Team is weakest, its violations a single repeated value rather than scatter (§6.1).

The prompt itself fails twice. The rubric contradicts itself — the innovation I-C band is 0.40–0.60 in the band list but 0.42–0.56 in the “AI-powered” cap — and of 42 I-C assignments all fall in the band range but only 45.2% in the cap range, clustering at the band endpoints: faced with the conflict, the model follows the band list. It also returned a category letter absent from the rubric (“S-D”) eight times, each accepted silently because the pipeline validates scores but not labels. Prompt-authored rules are therefore soft, model-specific constraints (§7.6), not guarantees: partially followed, silently declined, contradicted, or accepted malformed with nothing to catch it. Auditing them means verifying adherence in logged output rather than inferring it from prompt text — free once the pipeline persists per-stage artifacts (§3.1).

8 Limitations

Statistical. At $n = 12$, correlation tests remain underpowered ($|\rho| \approx 0.73$ required for 80% power): per-dimension estimates are wide and individually non-significant (Holm $p = 1.000$ throughout), and the three-rater panel (§6.1) establishes a reliability ceiling without increasing n . Of the two GO verdicts only one is stress-tested by repeated scoring: p7 sits within 0.008 of the 0.725 threshold across three runs and is fragile under the variability below (p3 fires via the unvalidated confidence branch, §3.4).

Methodological. Content-sensitivity checks are prompt-coupled, not independent tests of real-world responsiveness (§5.2). The groundedness audit (33 eligible claims) uses model-assisted, not double-annotated, labels. The rubric’s target band and blend weight ($\lambda = 0.90$) are documented heuristics (§3.4). Headline results use one provider (gpt-4o-mini); the cross-provider check (§7.6) replicates the expert-alignment failure but not its magnitude — one alternate model, not a family sweep, and unseeded. Stage 6’s scoring call is specified as deterministic (temperature= 0, seed= 42) but is not: across three runs on identical input, 11 of 12 proposals show at least one differing dimension score (overall MAD 0.0056; cells up to ± 0.12), concentrated in Objectives and Strategy. All 12 verdicts nonetheless held, weighted blending compressing the noise below the threshold distance — though only barely for p7. All reported results use a frozen canonical score set.

Domain and fairness. Kickstarter precision after recalibration remains modest (26–33%) — sufficient for triage, not automated decisions. Outcomes also depend on factors outside the proposal text (video, social following, marketing), capping text-only precision. Team scoring could still leak demographic signal beyond the gender axis tested, and the gender-name audit (§7.5) uses a single

constructed name pair rather than a validated corpus, with no sensitivity below its $n = 12$ /two-name resolution and no coverage of affiliation- or nationality-coded cues.

9 Conclusion and Future Work

We audited an evidence-grounded, nine-stage LLM venture-screening pipeline against a three-rater expert panel (12 proposals) and resolved funding outcomes for 141 Kickstarter campaigns. Useful outcome signal and decisive expert-alignment failure coexist: the score ranks unseen crowdfunding outcomes significantly, yet its agreement with panel consensus falls below every individual rater's leave-one-out agreement on every scorable target — a failure that replicates under a second provider while its magnitude does not, making the failure a property of the approach and its size a property of the model. Its decision threshold is separately invalid, flagging nothing out-of-domain until recalibrated, and its confidence does not track correctness. The question is therefore not whether an LLM can produce a plausible investment score, but what validation and governance must precede its use in accountable, human-supervised triage; nothing here justifies autonomous investment decisions or venture-success prediction.

The highest-priority next step is demographic-cue masking, with the fairness audit extended beyond a single constructed name pair. Style robustness and cross-dimension halo effects remain open; repeatability does not, though a fuller study would characterize the instability rather than establish it. Because Kickstarter supplies objective, resolved outcomes at scale, outcome-based validation of this form offers a practical route to auditing financial screening instruments before long-horizon labels mature.

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